Name:	Class:	Date:

Unit 1 Practical worksheet

Heating water with an immersion heater – the relationship between heat and temperature change

Science inquiry

- Investigate the relationship between heat added and temperature change for a fixed mass of water.

Context

Every time you boil a kettle or have a hot shower, electrical (or gas) energy is being transferred to water as heat. An electric kettle and an electric hot water system both use a heating element – a resistor that gets hot when current flows through it – sitting directly in the water. Energy companies, appliance designers and plumbers all need to predict how much energy it takes to raise the temperature of a given amount of water, and how long it will take. In this practical you will use a small version of a kettle element, an immersion heater, to add a measured amount of heat to a fixed mass of water and record how its temperature changes.

The heat added by an electrical heater can be calculated from the voltage across it, the current through it and the time it is switched on:

$$Q = VIt$$

where Q is the heat added (J), V is the voltage (V), I is the current (A) and t is the time (s).

Aim

To investigate whether the temperature change of water is proportional to the heat added, and to use the graph to determine a physical property of water.

Pre-lab questions

1 **Identify** the independent, dependent and controlled variables in this investigation. For each controlled variable, **explain** briefly why it must be kept the same.

Variable type	Variable	How it is measured or controlled
Independent		
Dependent		
Controlled 1		
Controlled 2		
Controlled 3		

2 **Predict** the shape of a graph of temperature change (vertical axis) against heat added (horizontal axis) if the two quantities are directly proportional. **Sketch** it in the space below.

Safety

⚠ CAUTION

- **Submerge the heater first.** The immersion heater must be fully under water BEFORE the power supply is switched on. **Never switch the heater on in air** – it will overheat within seconds, can burn out and can cause severe burns or a fire.
- **Switch off before removing.** Always turn the power supply off before lifting the heater out of the water. Hold it by the lead/handle only, and place it on a heatproof mat to cool. The metal casing stays hot for several minutes after switching off.
- **Hot water can scald.** Stop heating at 60 °C. Keep the cup inside a beaker so it cannot tip over, and do not carry the cup while it contains hot water.
- **Electricity and water.** Use only the low-voltage DC power supply provided – never mains power. Keep the power supply, meters and plugs well away from the water and work with dry hands. Check leads for damage before use. Wipe up spills immediately and tell your teacher.
- **Thermometers.** Do not use the thermometer as a stirrer. If a glass thermometer breaks, do not touch the glass – tell your teacher.
- Wear safety glasses and closed-in shoes throughout.

Materials

- 12 V immersion heater (approx. 50 W)
- Low-voltage DC power supply
- Digital voltmeter and digital ammeter, with connecting leads (OR a joulemeter)
- Insulated foam cup with a lid (hole for the heater and thermometer)
- 250 mL beaker (to hold the foam cup steady)
- Electronic balance (±0.01 g)
- Alcohol-in-glass thermometer (–10 to 110 °C, 1 °C divisions) OR digital temperature probe
- Stopwatch
- Stirrer
- Approximately 150 g of tap water per trial (three trials)
- Heatproof mat and paper towel

Method

1. Read the whole method and the safety box. Complete the pre-lab questions and fill in Table 1 (instrument uncertainties) before you begin.

2. Place the empty foam cup on the electronic balance and press **tare** (zero).
3. Add tap water to the cup until the balance reads approximately 150 g. Record the exact mass of water in Table 2. Use this same mass (as closely as possible) for all three trials.
4. Stand the foam cup inside the beaker for stability and place the lid on the cup.
5. Connect the circuit: the ammeter in **series** with the heater and the power supply, and the voltmeter in **parallel** (across) the heater. (If using a joulemeter, connect it between the power supply and the heater as instructed by your teacher.) Ask your teacher to check your circuit.
6. Lower the immersion heater through the lid so that the heating element is **fully submerged**. Do not switch on yet.
7. Insert the thermometer through the lid so that its bulb is in the water but **clear of the heater element** (not touching it).
8. Stir gently, then record the initial temperature of the water at t = 0 s in Table 3.
9. Set the power supply to 12 V. Switch on the power supply and start the stopwatch at the same moment.
10. Record the voltmeter and ammeter readings in Table 2. Check them again half-way through; if they change, record the average.
11. Stir gently and continuously throughout the heating. Record the temperature every 60 s for 8 minutes (480 s) in Table 3.
12. Switch off the power supply at 8 minutes **or** as soon as the water reaches 60 °C, whichever comes first.
13. With the power OFF, lift the heater out by its lead and place it on the heatproof mat. Carefully empty the warm water into the sink.
14. Allow the heater to cool in fresh water (power off) or wait until your teacher says it is ready. Repeat steps 2–13 for Trials 2 and 3 using **fresh room-temperature water** of the same mass each time.
15. Pack away equipment and wipe up any spills.

Set-up diagram

3 **Draw** a labelled diagram of your set-up, including the circuit (heater, power supply, voltmeter and ammeter), the foam cup, lid, thermometer and stirrer.

Results

Instrument uncertainties

Every measuring instrument has an **instrument uncertainty** (also called absolute uncertainty) that depends on its scale. Use these rules:

- **Analogue scale** (e.g. alcohol-in-glass thermometer, meter with a needle): uncertainty = **± half the smallest division**.
- **Digital scale** (e.g. digital balance, digital meter, stopwatch): uncertainty = **± the smallest increment** (the last digit shown).

Look closely at the instruments *you* are using and complete Table 1.

Table 1 Instrument uncertainties

Instrument	Analogue or digital?	Smallest division or increment (with unit)	Instrument uncertainty (± with unit)
Electronic balance			
Thermometer / temperature probe			
Voltmeter			
Ammeter			
Stopwatch			

4 The stopwatch displays time to 0.01 s. **Explain** why your reaction time may be a much larger source of uncertainty in the time measurements than the resolution of the stopwatch.

Raw data

Table 2 Mass of water and electrical readings

Trial	Mass of water (± _____ g)	Voltage *V* (± _____ V)	Current *I* (± _____ A)	Power *P* = *VI* (W)
1				
2				
3				
Mean				

If you are using a joulemeter, record the joulemeter reading (J) at each time in the margin of Table 3 instead of V and I, and use those readings as Q in Table 4.

Table 3 Temperature of the water during heating

Time *t* (± _____ s)	Trial 1 temperature (± _____ °C)	Trial 2 temperature (± _____ °C)	Trial 3 temperature (± _____ °C)	Mean temperature (°C)	Uncertainty in mean temperature (± °C) = ½ × range
0					
60					
120					
180					
240					
300					
360					
420					
480					

Uncertainty in the mean = (highest value – lowest value) ÷ 2. If this is smaller than the instrument uncertainty of the thermometer, use the instrument uncertainty instead.

Processed data

For each time, calculate the heat added using the **mean** voltage and **mean** current from Table 2, and the mean temperature change:

$Q = VIt$ ΔT = mean temperature at time t – mean temperature at t = 0 s

Table 4 Heat added and mean temperature change

Time *t* (s)	Heat added $Q = VIt$ (J)	Mean temperature change ΔT (°C)	Uncertainty in ΔT (± °C) (from Table 3)
0			
60			
120			
180			
240			
300			
360			
420			
480			

Worked example – calculating Q

Example values only – these are not real results. Use your own data in Table 4.

Suppose the mean voltage is V = 11.5 V, the mean current is I = 3.80 A, and we want the heat added at t = 180 s.

$Q = VIt$

Q = 11.5 V × 3.80 A × 180 s

Q = 7866 J

$Q = 7.87 \times 10^3$ J (3 significant figures, to match the data)

Note: 1 V × 1 A × 1 s = 1 W × 1 s = 1 J.

5 Using your own data, **calculate** Q for t = 240 s. Show all working and units.

Graphing

You will plot **mean temperature change ΔT (vertical axis)** against **heat added Q (horizontal axis)** as a scatter plot in Microsoft Excel. (Instructions are for Excel on Windows / Microsoft 365; menu names may differ slightly on a Mac.)

Enter the data

Step 1 Open a new Excel workbook. In cell A1 type **Heat added Q (J)**, in B1 type **Mean temperature change ΔT (°C)**, and in C1 type **Uncertainty in ΔT (°C)**.

Step 2 Enter your values from Table 4 in columns A, B and C, starting in row 2. (Enter the uncertainty as a positive number, without the ± sign.)

Create the scatter plot

Step 3 Select the data in columns **A and B only** (including the headings).

Step 4 Go to **Insert → Charts → Scatter** and choose **Scatter (markers only)**. Do not choose a version with lines joining the points.

Step 5 Check that Q is on the horizontal (x) axis and ΔT is on the vertical (y) axis. If they are the wrong way around, right-click the chart → **Select Data → Edit** and swap the ranges.

Titles and labels

Step 6 Click the chart, then click the **+ (Chart Elements)** button beside it. Tick **Axis Titles**.

Step 7 Click each axis title and type it **with units**: horizontal axis “Heat added, Q (J)”; vertical axis “Temperature change, ΔT (°C)”.

Step 8 Tick **Chart Title** and type a descriptive title, e.g. “Temperature change of 150 g of water versus heat added by an immersion heater”.

Step 9 Delete the legend (click it and press Delete) unless you have more than one data series.

Add error bars

Step 10 Click the chart, click **+ (Chart Elements)**, hover over **Error Bars** and click the arrow → **More Options…**

Step 11 Excel adds both horizontal and vertical error bars to scatter plots. Click the **horizontal** error bars on the chart and press **Delete** (you are only showing uncertainty in ΔT).

Step 12 Click the **vertical** error bars. In the *Format Error Bars* pane choose **Direction: Both**, **End Style: Cap**.

Step 13 Under **Error Amount**, choose **Custom → Specify Value**. For *Positive Error Value*, delete what is in the box and select your uncertainty values in column C (C2 down to the last row). Do the same for *Negative Error Value*. Click **OK**.

Add a linear trendline

Step 14 Right-click any data point → **Add Trendline…**

Step 15 In the *Format Trendline* pane choose **Linear**.

Step 16 Tick **Display Equation on chart** and **Display R-squared value on chart**. Do **not** tick *Set Intercept* – you need Excel to calculate the intercept for you.

Step 17 Drag the equation box to a clear area and increase its font size so it is easy to read. Change the number format if needed so the gradient shows at least 3 significant figures (right-click the equation → **Format Trendline Label → Number → Scientific**, 2 decimal places).

Deciding which points to include

When the heater is first switched on, some of the heat goes into warming the heater itself, the foam cup and the thermometer before the water temperature rises steadily. This can make the first one or two points sit away from the straight-line pattern.

Step 18 Look at your graph. If the first point(s) clearly do not follow the trend of the rest, you may exclude them **from the trendline only** (still show them on the graph). To do this: right-click the chart → **Select Data** → **Add**. Name the new series "Points used for trendline", and select only the Q and ΔT values you are keeping. Delete the trendline from the original series and add it to the new series instead (Steps 14–17).

Step 19 Copy your finished graph and paste it into the box below (or print and attach it).

Paste or attach your graph here

6 **Justify** your decision to include or exclude the early data points from your trendline. Refer to the position of the points on your graph and to what is physically happening during the first minute of heating.

Analysis

7 From your trendline equation, **identify** the following. Include units for each (use the units on your axes).

Quantity	Value (with units)
Gradient	
y-intercept	
R^2 value	

8 **Describe** the trend shown in your graph. Does your graph show that ΔT is directly proportional to Q? **Justify** your answer by referring to the shape of the graph, the R^2 value and the *y*-intercept.

Interpreting the gradient

The heat needed to change the temperature of a substance is given by:

$$Q = mc\Delta T$$

where Q is the heat added (J), m is the mass (kg), c is the specific heat capacity of the substance (J kg^{-1} K^{-1}) and ΔT is the temperature change (K or °C). Note that a temperature *change* of 1 °C is the same size as a change of 1 K.

A straight-line graph has the equation **y = mx + c**. (Be careful – the m and c in $y = mx + c$ are **not** the same as the mass m and specific heat capacity c!)

9 Work through the following steps to **infer** what physical quantity the gradient of your graph represents.

(a) Which variable is on your *y*-axis? Which variable is on your *x*-axis?

y-axis variable: _____ *x*-axis variable: _____

(b) **Rearrange** $Q = mc\Delta T$ so that the *y*-axis variable is the subject. Show each step.

(c) Now line your rearranged equation up with $y = mx + c$. Once you have matched y and x, what is left over? Complete the table.

In $y = mx + c$	In my rearranged equation
y	
x	
gradient (m)	
y-intercept (c)	

(d) **Infer** what physical quantity (or combination of quantities) the gradient of your graph represents. **Explain** how you know.

(e) **Show** that the units of your answer to (d) are consistent with the units of your gradient from Question 7.

Determining a physical property of water

10 Use your gradient and your measured mass of water (in **kg**) to **calculate** the specific heat capacity of water, c. Show all working, give your answer to an appropriate number of significant figures and include units (J kg^{-1} K^{-1}).

11 The accepted value for the specific heat capacity of water is **4180 J kg^{-1} K^{-1}**. **Calculate** the absolute error and the percentage error in your experimental value.

Absolute error = | experimental value – accepted value |

Percentage error = (absolute error ÷ accepted value) × 100%

Interpreting the intercept

12 **Infer** what the *y*-intercept of the graph should be in theory. **Explain** your reasoning using the physics of the situation (what happens to ΔT when $Q = 0$ J?).

13 **Identify** the value of your actual *y*-intercept and **explain** what it means physically. If it is not zero, suggest why.

Evaluation

Reliability – the consistency of results when the experiment is repeated. Reliability relates to **random error** and **precision**. Random errors cause measurements to scatter unpredictably above and below the true value. If repeated trials give very similar results (a small spread) and the data points sit close to the line of best fit, the results are reliable (precise).

Validity – whether the method actually measures what the aim intends it to measure. Validity relates to **systematic error** and **accuracy**. Systematic errors shift every measurement in the same direction (all too high or all too low) and cannot be reduced by doing more trials. A valid method produces results close to the true (accepted) value.

Reliability

14 Use the **spread between your three trials** (the uncertainty column in Table 3) and your **R^2 value** to **identify** and **explain** limitations that affected the reliability of your results.

(a) **Identify** the largest uncertainty in the mean temperature in Table 3. At which part of the heating did it occur? **Suggest** why.

(b) **Explain** what your R^2 value tells you about the scatter of your data about the trendline, and therefore about the reliability of your results.

(c) **Identify** two sources of random error in this experiment and **explain** how each one would affect your readings.

Validity

15 The method assumes that **all** of the electrical energy supplied ($Q = VIt$) goes into heating the water.

(a) **Identify** two ways in which some of the electrical energy supplied does **not** end up raising the temperature of the water.

(b) **Explain** how each limitation in (a) affects the measured temperature change ΔT for a given amount of heat supplied.

(c) **Infer** whether your experimental value of c would be **too high** or **too low** as a result. **Justify** your answer using the equation $Q = mc\Delta T$ (rearranged to make c the subject). Does your percentage error from Question 11 support your inference?

(d) **Identify** one other limitation of the method that could affect validity (think about where the thermometer is, how well the water is mixed, or whether V and I stayed constant).

Classifying errors

16 List each limitation you identified in Questions 14 and 15 in Table 5. **Classify** each as a random or systematic error, and state whether it mainly affects reliability or validity.

Table 5 Classification of limitations

Limitation	Random or systematic error?	Mainly affects reliability or validity?

Improvements

17 **Propose** one improvement for **each** limitation in Table 5. Your improvement must be specific and realistic for a school laboratory. Structure each answer using the headings in Table 6.

Table 6 Proposed improvements

The change suggested	The problem it addresses	How the change addresses the problem

Conclusion

18 **Draw a conclusion** about the aim of this investigation that is **justified** by your data. Your conclusion should:

- state whether the temperature change of water is directly proportional to the heat added, referring to your graph (shape, R^2 and y-intercept)
- state your experimental value of the specific heat capacity of water, with its percentage error
- comment on how much confidence you have in your result, based on your evaluation.